Arjun Chavan

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EDUCATION

University of Michigan, Ann Arbor, MI — Expected Dec 2027
Bachelor of Engineering in Computer Science
Relevant Coursework: Data Structures and Algorithms, Computer Vision, Computer Architecture, Applied Linear Algebra, Discrete Mathematics; Machine Learning, Embedded Systems, and Autonomous Robotics (In Progress)

SKILLS

Languages: Python, C, C++, Java, SQL, JavaScript, TypeScript, HTML, CSS, Bash, LaTeX, Verilog, Assembly, R
Frameworks and Tools: React, Vite, Node.js, REST APIs, Git, GitHub, Linux, Unix, VS Code, Figma, Agile, Scrum
AI and ML: OpenCV, PyTorch, TensorFlow, Prompt Engineering
Hardware: Circuit Design, Arduino, Raspberry Pi, FPGAs, Logic Analyzers, Magnetic Induction, Soldering

EXPERIENCE

Tech Lead, Altheros Capital, Ann Arbor, MI — May 2026 – Oct 2026
- Co-led a 10-engineer fullstack team on Midwest Health, a healthcare patient-scheduling platform for an Altheros Capital portfolio company, planning sprints
- Reviewed pull requests and held merge authority for the team, gating all changes before production deployment
- Presented monthly product demos to a 60-person audience including an Altheros Capital partner
- Scoped a two-week MVP timeline and two-developer staffing plan for Nerd Perfect Media, recommendation adopted
- Led a schema audit, facilitating cross-team discussion to align frontend and backend on a shared data format

Full Stack Developer Intern, Altheros Capital, Ann Arbor, MI — Feb 2026 – Aug 2026
- Shipped React landing pages and UI components for Midwest Health and Nerd Perfect Media from Figma designs
- Collaborated in Agile and Scrum sprint planning, translating user stories into actionable UI tasks
- Migrated build tooling from deprecated react-scripts to Vite, restoring CI pipeline stability
- Authored the team deployment runbook covering release packaging, secure upload, and rollback procedures

Machine Learning Research Assistant, UMass Lowell (PeARL Lab), Lowell, MA — May 2023 – Aug 2023
- Prototyped machine learning models for robotic arm pathfinding and manipulation, improving task success rates
- Automated data collection pipelines in Python, reducing preparation time for the research team
- Deployed control scripts that increased the success rate of robotic pick-and-place operations

Machine Learning Engineer, MIT Lincoln Laboratory (Beaver Works), Cambridge, MA — July 2023 – Aug 2023
- Implemented autonomous navigation using OpenCV to track pool lane lines in real time
- Processed AprilTag detection data to compute relative 3D position and heading between robotic units
- Tuned PID controllers to stabilize robot orientation, producing smoother and more consistent trajectory tracking

PROJECTS

A* Path Planner and Robotics Runtime, Python, TCP, JSON, EECS 367 — Sept 2026
- Recreated ROS publish and subscribe semantics without ROS libraries, serving clients over a TCP and JSON gateway
- Planned optimal 4-connected paths with A* over occupancy grids, backed by a hand-written binary min-heap
- Scored 100% on a black-box autograder validating paths against unseen staff-authored grading maps

Forward Kinematics for URDF Robots, Python, TCP, JSON, EECS 367 — Oct 2026
- Wrote a URDF parser and forward-kinematics solver from scratch, composing joint transforms into link poses
- Published link transforms at 10 Hz over a TCP and JSON pub/sub gateway, rejecting malformed robot descriptions safely
- Verified with 116 unit and wire-level tests, and hand-choreographed a dual-arm demo robot dance in the live viewer

ICU Mortality Prediction, Python, scikit-learn, EECS 445 — Oct 2026
- Expanded 40 max-only ICU features into 253 trend and measurement-count features across 10,000 patients
- Tuned logistic regression across 20 settings with stratified 5-fold cross-validation, reaching 0.858 AUROC
- Raised out-of-fold F1 from 0.517 to 0.551 by tuning the decision threshold on out-of-fold scores

Gamified Financial Dashboard, React, Expo, Node.js, TypeScript, CMU TartanHacks — Feb 2026
- Created a gamified personal finance dashboard with an AI-driven virtual pet system and real-time quest overlays
- Developed a TypeScript onboarding engine parsing the Capital One REST API to surface AI-curated financial insights
- Secured 1st Place in the Visa Track, presenting a scalable fintech solution to industry executives